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Roll# 2022F-BCS-152
Section: C

Compiler Construction Assignment #3

Q) By Applying the algorithm of Non Recursive Predictive Parser & SLR Parser to draw both of the parser by using the Given Grammar.

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow id$

This grammar is left recursive, which is not suitable for a predictive parser. We must first eliminate left recursion.

Eliminate Left Recursion

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \epsilon$

$F \rightarrow id$

First $(E) = \{id\}$	Follow $(E) = \{\$\}$
First $(T) = \{id\}$	Follow $(E') = \{\$\}$
First $(F) = \{id\}$	Follow $(T) = \{+, \$\}$
First $(E') = \{+, \epsilon\}$	Follow $(T') = \{+, \$\}$
First $(T') = \{*, \epsilon\}$	Follow $(F) = \{*, +, \$\}$

Parsing table

Non Terminal	id	+	*	$
E	TE'			
E'		+TE'		ε
T	FT'			
T'		ε	*FT'	ε
F	id			

Input : id + id * id $

Stack : $ E (Start symbol and end marker)

Stack	Input	Output
$E	id + id * id $	$E \rightarrow TC'$
$ E'T	id + id * id $	$T \rightarrow FT'$
$ E'T'F	id + id * id $	$F \rightarrow id$
$ E'T'	+ id * id $	$T' \rightarrow ε$
$ E'	+ id * id $	$E' \rightarrow +TE'$
$ E'T +	id * id $	$T \rightarrow FT'$
$ E'T' *F +	id * id $	$F \rightarrow id$
$ E'T' +	* id $	$T' \rightarrow *FT'$
$ E'T' F * +	id $	$F \rightarrow id$
$ E'T' * +	$	$T' \rightarrow ε$
$ E' +	$	$E' \rightarrow ε$
	$	Accept

~~SP~~ SLR Parsing.

I_0:

$E' \rightarrow .E$

$E \rightarrow .E+T$

$E \rightarrow .T$

$T \rightarrow .T*F$

$T \rightarrow .F$

$F \rightarrow .id$

Goto (I_0, E)

$$\left[\begin{array}{l} E' \rightarrow E. \\ E \rightarrow E.+T \end{array}\right] I_1$$

goto (I_0, T)

$$\left[\begin{array}{l} E \rightarrow T. \\ T \rightarrow T.*F \end{array}\right] I_2$$

goto (I_0, F)

$$\left[T \rightarrow F. \right] I_3$$

goto (I_0, id)

$$\left[F \rightarrow id. \right] I_4$$

goto $(I_1, +)$

$$\left[\begin{array}{l} E \rightarrow E+.T \\ T \rightarrow .T*F \\ T \rightarrow .F \\ F \rightarrow .id \end{array}\right] I_5$$

goto $(I_2, *)$

$$\left[\begin{array}{l} T \rightarrow T*.F \\ F \rightarrow .id \end{array}\right] I_6$$

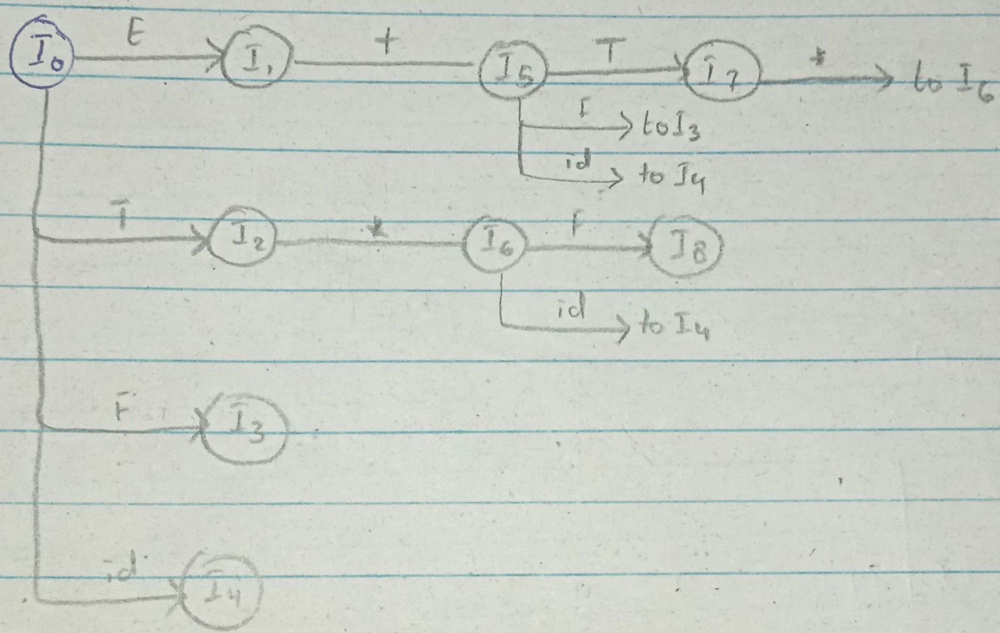
goto (I_5, T)

$$\left[\begin{array}{l} E \rightarrow E+T. \\ T \rightarrow T.*F \end{array}\right] I_7$$

goto (I_6, F)

$[T \to T*F.]_{I_8}$

Follow (E') = {$}
Follow (E) = {$, +}
Follow (T) = {$, +, *}
Follow (F) = {$, +, *}



State	id	+	*	$	E	T	F
0	s4				1	2	3
1		s5		acc			
2		r2	s6	r2			
3		r4	r4	r4			
4		r5	r5	r5			
5	s4					7	3
6	s4						8
7		r1	s6	r1			
8		r3	r3	r3			